1) correlation & covariance

X	Y
1	2
2	4
3	5
4	4
5	5

X: Hours studied

Y: Exam score

Correlation

$$r = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

$$\text{covarience (cov)} = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

$$\sigma_x = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}} \qquad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x^{(i)}$$

$\bar{x} = \frac{15}{5} = 3$ $\quad \bar{y} = \frac{20}{5} = 4$

$\sigma_x = \sqrt{\frac{10}{5}}$ $\quad \sigma_y = \sqrt{6/5}$

$\sigma_x = \sqrt{2}$ $\quad \sigma_y = \sqrt{1.2}$

$\sigma_x = 1.414$ $\quad \sigma_y = 1.09$

x	$x-\bar{x}$	$(x-\bar{x})^2$
1	-2	4
2	-1	1
3	0	0
4	1	1
5	2	4

y	$y-\bar{y}$	$(y-\bar{y})^2$
2	-2	4
4	0	0
5	1	1
4	0	0
5	1	1

$$cov = \frac{1}{5}\Big[(1-3)(2-4) + (2-3)(4-4) + (3-3)(5-4) + (4-3)(4-4) + (5-3)(5-4)\Big]$$

$$= \frac{1}{5}\Big[(-2)(-2) + (2)(1)\Big]$$

$$= \frac{1}{5}[4+2]$$

$$= 1.2$$

$$r = \frac{1.2}{(1.414)(1.09)}$$

$$= \frac{1.2}{1.541}$$

$$= 0.77$$

Eigen value

$$\det(A - \lambda I) = 0$$

$$\begin{array}{c|c} & 3 \\ \hline -3 & -4 \\ \hline & -4 \end{array}$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = 0$$

$$(2-\lambda)(2-\lambda) - 1 = 0$$

$$4 - 4\lambda + \lambda^2 = 0$$

$$\lambda^2 - 4\lambda + 4 - 1 = 0$$

$$\boxed{\lambda_1 = 3, 1} \quad \Rightarrow \lambda_1 = 3, \quad \lambda_2 = 1$$

<u>Compute eigenvectors.</u>

For $\lambda_1 = 3$

$$u = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(A - \lambda I) u = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} u = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-x_1 + x_2 = 0 \Rightarrow x_1 = x_2$$

If $x_1 = 1$,

$$u_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\lambda = 1$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

If $x_1 = 1$

$x_2 = -1$

PCA

$$X = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \qquad d = 3$$

$$\mu = \begin{bmatrix} 1.5 & 1.5 & 1.5 \end{bmatrix}$$

$$X_c = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix}$$

Covariance matrix

$$\begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) & cov(x_1, x_3) \\ cov(x_2, x_1) & cov(x_2, x_2) & cov(x_2, x_3) \\ cov(x_3, x_1) & cov(x_3, x_2) & cov(x_3, x_3) \end{bmatrix}$$

$$C = \frac{1}{m} X_c^T X_c \Rightarrow \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix}$$

Eigen Value &

Eigen Vectors

$\lambda_1 = 2.0, \quad \lambda_2 \approx 0.75, \quad \lambda_3 \approx 0.0$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Dimension reduction

3D → 2D

Take top 2 eigen vector

$$W_{2D} = [v_1 \ v_2]$$

Project the data point

$$Z = X_c W_{2D}$$

Each point will be of 2D

Take top 1 eigen vector

$$W_{1D} = [v_1]$$

Total variance

$$= 2 + 0.75 + 0$$

$$= 2.75$$

$$PC_1 = \frac{2}{2.75} \approx 73\%$$

$$PC_2 = \frac{0.75}{2.75} \approx 27\%$$

$$PC_3 = \frac{0}{2.75} \approx 0\%$$